The streamline upwind Petrov-Galerkin method (SUPG) and the Galerkin finite element method for convection-diffusion equations

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Abstract

In this paper, I will examine the FEM solution for convection-dominated convection-diffusion equations. First, Galerkin's finite element method will be applied for different diffusivity constants, and the drawbacks will be demonstrated through numerical studies. Second, to address the problem, the SUPG method will be proposed, and the same problem will be solved using this method.

1. Problem formulation

Throughout this paper, we consider the following convection-dominated convection-diffusion equation with a nonhomogeneous Dirichlet boundary condition:

$$\begin{aligned} -\varepsilon\Delta\bar{u} + b\cdot\nabla\bar{u} &= f \qquad \text{in } \Omega, \\ \bar{u} &= u_b \qquad \text{on } \partial\Omega, \end{aligned} \tag{1}$$

where $\Omega \subset \mathbb{R}^2$ (resp. $\mathbb{R}^3$) is a polygonal domain (a polyhedral domain), ε is a positive diffusivity constant, $b \in W^{1,\infty}(\Omega)^2$ (resp. $W^{1,\infty}(\Omega)^3$) is the convection coefficient with the incompressibility condition $\nabla\cdot b = 0$, $f \in L^2(\Omega)$ is a source function, and $u_b \in H^{1/2}(\partial\Omega)$ is the Dirichlet boundary condition.

Let $\bar{u}$ be the solution of the problem (1) such that $\bar{u} = u + u_p$, where u is the solution of the corresponding homogeneous problem and u_p is a particular solution. The trace of u_p on $\partial\Omega$ is the function u_b, and the function u_p is called a Dirichlet lift of u_b in Ω. Then u is the solution of the following homogeneous problem:

$$\begin{aligned} -\varepsilon\Delta u + b\cdot\nabla u &= f + \varepsilon\Delta u_p - b\cdot\nabla u_p \qquad \text{in } \Omega, \\ u &= 0 \qquad \text{on } \partial\Omega. \end{aligned}$$

2. The Galerkin Finite Element Method (GFEM)

Let us find a weak (or variational) formulation of the convection-diffusion equation. Define a variational space that incorporates the essential (i.e., Dirichlet) boundary condition.

$$V := H_0^1(\Omega) = \{v \in H^1(\Omega) : v|_{\partial\Omega} = 0\}.$$

Multiply both sides of the equation by a test function $v \in V$ and integrate over the domain Ω.

$$\int_\Omega (\varepsilon(\Delta\bar{u})v + (b\cdot\nabla\bar{u})v)\,dx = \int_\Omega fv\,dx.$$

Once the Green's first identity is used, we obtain

$$\varepsilon \int_\Omega \nabla \bar{u} \cdot \nabla v \, dx - \varepsilon \int_{\partial\Omega} \frac{\partial \bar{u}}{\partial \vec{n}} v \, ds + \int_\Omega (b \cdot \nabla \bar{u}) v \, dx = \int_\Omega f v \, dx.$$

Since $v \in H_0^1(\Omega)$, we have

$$\varepsilon \int_\Omega \nabla \bar{u} \cdot \nabla v \, dx + \int_\Omega (b \cdot \nabla \bar{u}) v \, dx = \int_\Omega f v \, dx,$$

or equivalently,

$$\varepsilon \int_\Omega \nabla u \cdot \nabla v \, dx + \int_\Omega (b \cdot \nabla u) v \, dx = \int_\Omega f v \, dx - \varepsilon \int_\Omega \nabla u_p \cdot \nabla v \, dx - \int_\Omega (b \cdot \nabla u_p) v \, dx,$$

where $u_p \in H^1(\Omega)$. Then, the weak formulation of the problem is as follows: find $u \in V$ such that

$$a(u, v) = l(v), \quad \forall v \in V, \tag{2}$$

where

$$\begin{aligned} a(u, v) &= \varepsilon \int_\Omega \nabla u \cdot \nabla v \, dx + \int_\Omega (b \cdot \nabla u) v \, dx = \varepsilon(\nabla u, \nabla v) + (b \cdot \nabla u, v), \\ l(v) &= \int_\Omega f v \, dx - \varepsilon \int_\Omega \nabla u_p \cdot \nabla v \, dx - \int_\Omega (b \cdot \nabla u_p) v \, dx = (f, v) - a(u_p, v). \end{aligned}$$

Let us now verify that there exists a unique weak solution $u \in H^1(\Omega)$.

Theorem 1. *There exists a unique $u \in V$ satisfying* (2).

Proof. The variational space V is a Hilbert space with the inner product

$$(u, v)_{H^1(\Omega)} = \int_\Omega uv \, dx + \int_\Omega \nabla u \cdot \nabla v \, dx.$$

Now, let us show that $l : V \to \mathbb{R}$ is a continuous linear functional. For any $\alpha, \beta \in \mathbb{R}$ and $v_1, v_2 \in V$, we have

$$\begin{aligned} l(\alpha v_1 + \beta v_2) = (f, \alpha v_1 + \beta v_2) - a(u_p, \alpha v_1 + \beta v_2) = \alpha(f, v_1) + \beta(f, v_2) - \\ - \alpha a(u_p, v_1) - \beta a(u_p, v_2) = \alpha l(v_1) + \beta l(v_2), \end{aligned}$$

which implies that $l(\cdot)$ is a linear functional. As $l(\cdot)$ is a linear functional, showing its boundedness suffices to establish continuity. Using the Cauchy-Schwarz inequality, the continuity of $a(\cdot, \cdot)$, and the Sobolev embedding theorem, we get the following inequality.

$$\begin{aligned} |l(v)| = \left| \int_\Omega (f, v) - a(u_p, v) \right| \leq \left| \int_\Omega (f, v) \right| + |a(u_p, v)| \leq \|f\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} + \|u_p\|_{H^1(\Omega)} \|v\|_{H^1(\Omega)} \\ \leq \left(\|f\|_{L^2(\Omega)} + \|u_p\|_{H^1(\Omega)} \right) \|v\|_{H^1(\Omega)} = M \|v\|_{H^1(\Omega)}, \end{aligned}$$

where $M = \|f\|_{L^2(\Omega)} + \|u_p\|_{H^1(\Omega)} > 0$. Since $l(\cdot)$ is found to be bounded, it is continuous.

Next, we will show that $a : V \times V \to \mathbb{R}$ is a continuous and coercive bilinear form. Notice that for any fixed $w \in V$, $a(w, v)$ is linear for all $v \in V$. Similarly, for any fixed $w \in V$, $a(u, w)$ is linear for all $u \in V$. So, $a(\cdot, \cdot)$ is a bilinear functional on $V \times V$.

$$\begin{aligned} |a(u, v)| = \left| \varepsilon \int_\Omega \nabla u \cdot \nabla v \, dx + \int_\Omega (b \cdot \nabla u) v \, dx \right| \leq \varepsilon \left| \int_\Omega \nabla u \cdot \nabla v \, dx \right| + \left| \int_\Omega (b \cdot \nabla u) v \, dx \right| \leq \\ \leq \varepsilon \|\nabla u\|_{L^2(\Omega)} \|\nabla v\|_{L^2(\Omega)} + \|b \cdot \nabla u\|_{L^2(\Omega)} \|v\|_{L^2(\Omega)} \end{aligned}$$

To show that a bilinear form $a(\cdot,\cdot)$ is coercive on V, we need to find a positive constant c_{coer} such that $a(v,v) \geq c_{\text{coer}}\|v\|_V^2$ for all $v \in V$.

$$a(v,v) = \varepsilon \int_\Omega \nabla v \cdot \nabla v \, dx + \int_\Omega b \cdot (\nabla v) v \, dx = \varepsilon |\nabla v|_{H^1(\Omega)} + \int_\Omega b \cdot (\nabla v) v \, dx.$$

To proceed further, notice that $v(\nabla v) = \frac{1}{2}\nabla(v^2)$ and use integration by parts to obtain the following equality.

$$\begin{aligned} a(v,v) &= \varepsilon |v|_{H^1(\Omega)}^2 + \frac{1}{2}\int_\Omega b \cdot \nabla(v^2) \, dx = \\ &= \varepsilon |v|_{H^1(\Omega)}^2 + \frac{1}{2}\int_{\partial\Omega} v^2 (b \cdot \vec{n}) \, ds - \frac{1}{2}\int_\Omega v^2 (\nabla \cdot b) \, dx. \end{aligned}$$

Since $v \in H_0^1(\Omega)$ and $\nabla \cdot b = 0$ (incompressibility condition), we have

$$a(v,v) = \varepsilon |v|_{H^1(\Omega)}^2.$$

Once the Poincaré inequality is applied, we obtain

$$a(v,v) = \varepsilon |v|_{H^1(\Omega)}^2 = \frac{\varepsilon}{2}|v|_{H^1(\Omega)}^2 + \frac{\varepsilon}{2}|v|_{H^1(\Omega)}^2 \geq \frac{\varepsilon}{2}|v|_{H^1(\Omega)}^2 + \frac{\varepsilon}{2c_{\text{P}}^2}\|v\|_{L^2(\Omega)}^2 \geq c_{\text{coer}}\|v\|_{H^1(\Omega)}^2, \quad (3)$$

where $c_{\text{coer}} = \frac{\varepsilon}{2+2c_{\text{P}}^2}$. The inequality (3) shows that $a(\cdot,\cdot)$ is coercive.

Since all conditions of the Lax-Milgram theorem are satisfied, then there exists a unique $u \in V$ satisfying (2). □

Let V_h be a finite dimensional subspace of V. Then, the Galerkin finite element discretization of (1) is as follows: find $u_h \in V_h$ such that

$$a(u_h, v_h) = l(v_h), \quad \forall v_h \in V_h. \quad (4)$$

3. The Streamline upwind Petrov-Galerkin Method (SUPG)

For convection-dominated problems, the Péclet number is very large. For the problem (1), it is given by

$$Pe = \frac{1}{\varepsilon}\left(\sum_{i=1}^n \|b_i\|_{L^\infty(\Omega)}^2\right)^{1/2} \quad (5)$$

$$a_h(u_h, v_h) + (R_h(u_h), \tau b \cdot \nabla_h v_h) = (f, v_h)$$

$R_h(u) = -\varepsilon \Delta_h u + b \cdot \nabla_h u - f$

$$\tau = \frac{h}{2|b|}\xi_0(Pe) \text{ with } \xi_0(\alpha) = \coth\alpha - \frac{1}{\alpha}, Pe = \frac{|b|h}{2\varepsilon}$$

$$\tau = \frac{h}{2|b|}\xi_1(Pe) \text{ with } \xi_1(\alpha) = \max\{0, 1 - \frac{1}{\alpha}\}, Pe = \frac{|b|h}{2\varepsilon}$$

$$\tau = \frac{h}{2|b|}\xi_1(Pe) \text{ with } \xi_2(\alpha) = \min\{1, \frac{\alpha}{3}\}, Pe = \frac{|b|h}{2\varepsilon}$$

$$\tau_K = \frac{diam(K)}{2|b|}\xi_0(Pe_K/2) \text{ with } \xi_0(\alpha) = \coth\alpha - \frac{1}{\alpha}, Pe_K = \frac{|b| diam(K)}{2\varepsilon}, diam(K) = \sup\{|x-y| : x, y \in K\}$$

$$\tau_K = \begin{cases} \tau_0 h_K, & \text{if } Pe_K > 1, \\ \tau_1 h_K^2/\varepsilon, & \text{if } Pe_K \leq 1, \end{cases}$$

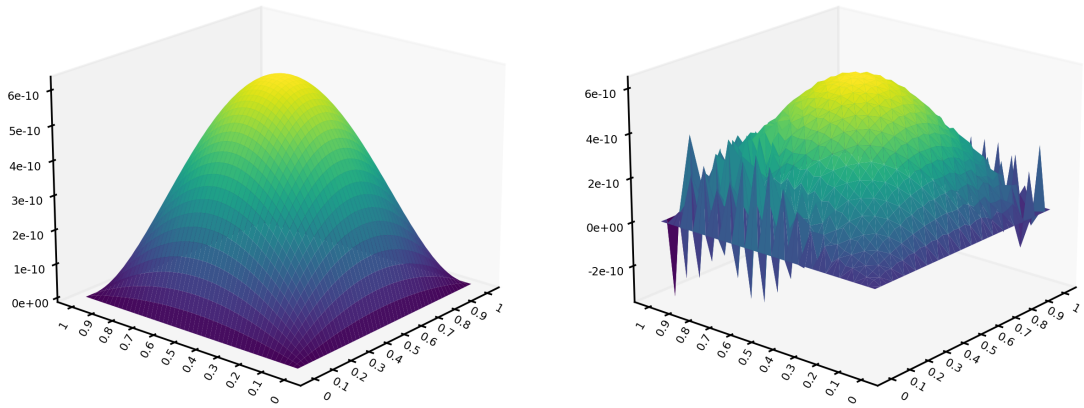
4. Numerical experiments

Example 1. Consider the following convection-diffusion equation

$$\begin{aligned} -\varepsilon\Delta u + b\cdot\nabla u &= f \qquad \text{in } (0,1)\times(0,1), \\ u(x,y) &= 0 \qquad \text{on } [0,1]\times[0,1], \end{aligned} \tag{6}$$

where $\varepsilon = 10^{-8}$, $b = (5,3)^{\mathrm{T}}$ and $f = 2\varepsilon^2\left(y(1-y)+x(1-x)\right) + 5\varepsilon y(1-y)(1-2x) + 3\varepsilon x(1-x)(1-2y)$. The exact solution of the problem is $u(x,y) = \varepsilon xy(1-x)(1-y)$.

Once we use the Galerkin Finite Element Method (GFEM) to solve the problem, we obtain 1b. It is clear to see that



(a) Exact solution.

(b) GFEM solution for $n = 1089$.

Figure 1

n	$\|\|e_h\|\|_{L^2}$	L^2 rate	$\|\|e_h\|\|_{H^1}$	H^1 rate	$\|\|e_h\|\|_{L^\infty}$	L^∞ rate
25	73655.7941	2.00002	836581.8743	1.0166	125009.6760	1.9999
81	18413.3007	2.00005	417073.3502	1.0042	31277.3595	1.9988
289	4754.7041	1.95332	212826.6730	0.9706	9666.3031	1.6941
1089	1226.6493	1.95463	108607.1038	0.9706	2484.1977	1.9602
4225	309.7504	1.98554	54725.6699	0.9888	641.1749	1.9540
16641	77.4897	1.99903	27515.4558	0.9920	163.7736	1.9690

Example 2. Consider the following convection-diffusion equation

$$\begin{aligned} -\varepsilon\Delta u + b\cdot\nabla u &= f \qquad \text{in } (0,1)\times(0,1), \\ u(x,y) &= 0 \qquad \text{on } [0,1]\times[0,1], \end{aligned} \tag{7}$$

where $\varepsilon = 10^{-6}$, $b = (\cos(-\pi/3), \sin(-\pi/3))^{\mathrm{T}}$ and $f = -2\varepsilon^2\pi^2\sin(\pi x)\sin(\pi y)(x-0.5)(y-0.5) + 4\varepsilon^2\pi\sin(\pi x)\sin(\pi y) + \frac{1}{2}\left(\varepsilon\pi\cos(\pi x)\sin(\pi y)(x-0.5)(y-0.5) + \varepsilon\sin(\pi x)\sin(\pi y)(y-0.5)\right) - \frac{\sqrt{3}}{2}\left(\varepsilon\sin(\pi x)\pi\cos(\pi y)(x-0.5)(y-0.5) + \varepsilon\sin(\pi x)\sin(\pi y)(x-0.5)\right)$. The exact solution of the problem is $u(x,y) = \varepsilon\sin(\pi x)\sin(\pi y)(x-0.5)(y-0.5)$.

References

[1] S. C. Brenner and L. R. Scott, *The Mathematical Theory of Finite Element Methods*, 3rd ed., Springer, 2008. DOI: 10.1007/978-0-387-75934-0.